



3500 Cylinder Head Water Passage Insert Fixture

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Insert Enabled with Inclined Holes for Heads 3500	
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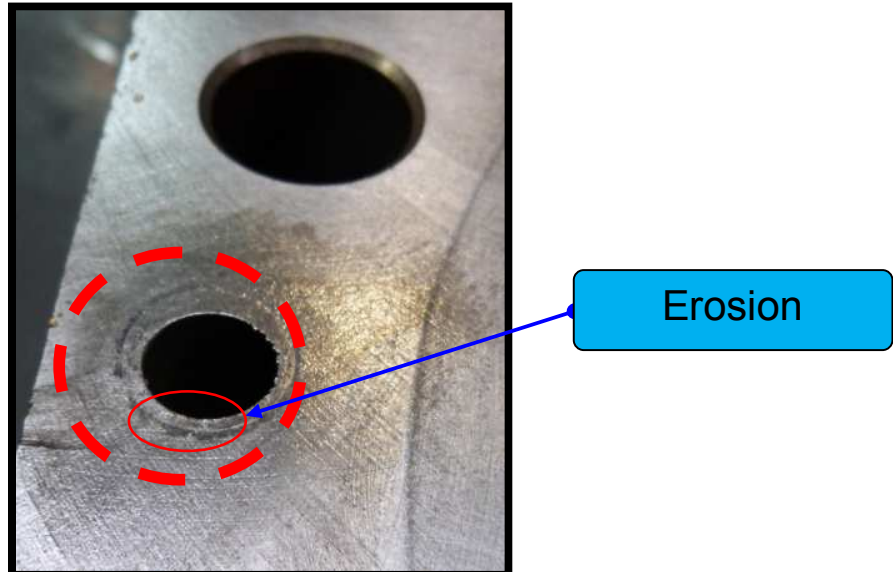
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1.0 Introduction

A new fixture for machining water passage inserts has been implemented at Ferreyros Machine Shop, reducing labor time.

2.0 Best Practice Description

In the lateral water passages, where the hole is angled, erosion around the passage has been seen.

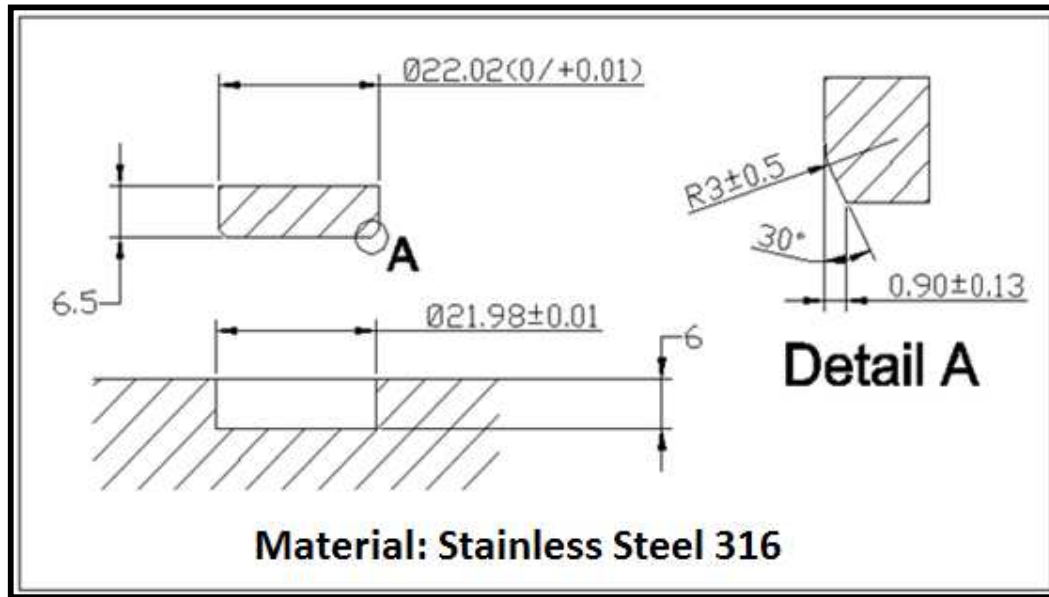


Picture 2: Detachment in cleaning metallic

3.0 Implementation Steps

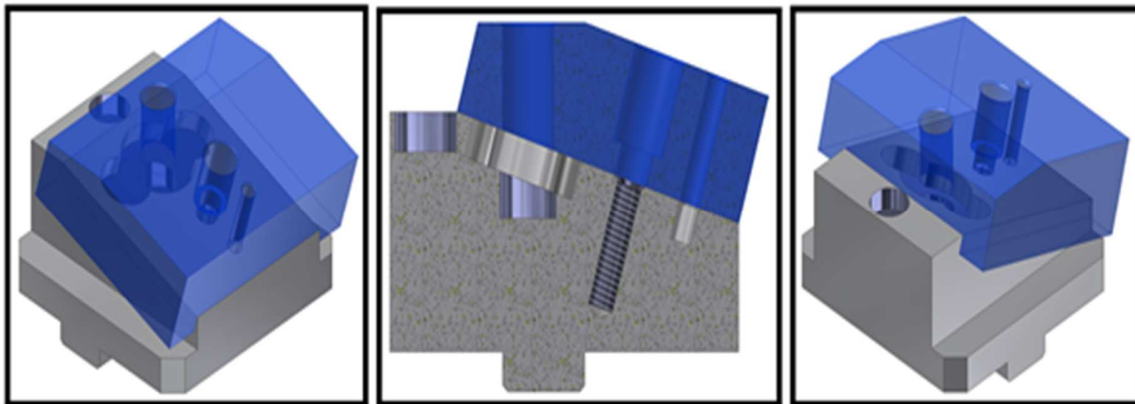
Blank inserts are initially machined on a CNC machine to the following specifications (see picture 2).

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Picture 3: Dimensions of the insert pass water

For this procedure a device was prepared (See Picture 4) in which the inserts were drilled for the 17° as the technical information requested.



Picture 4: Device for head's inserts

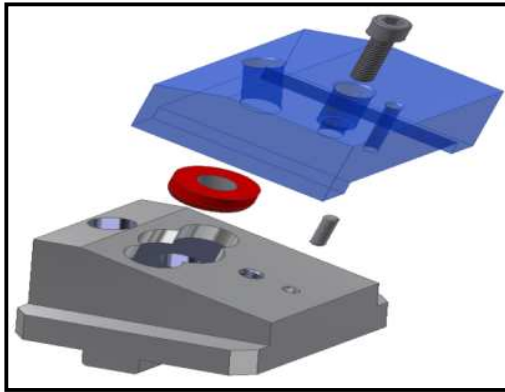
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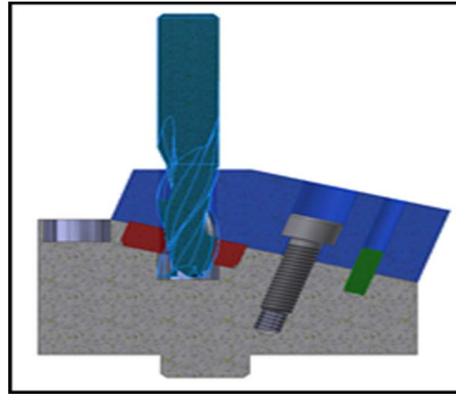
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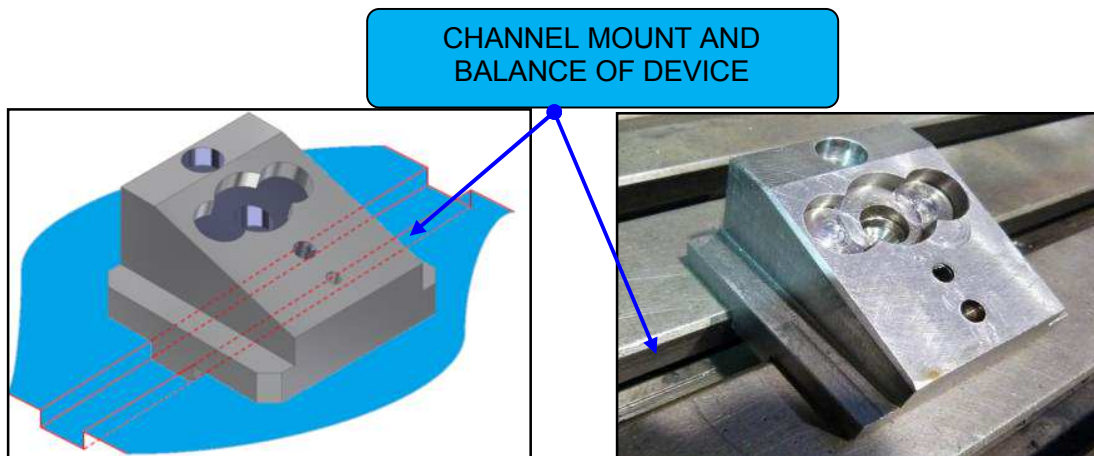


Isometric View



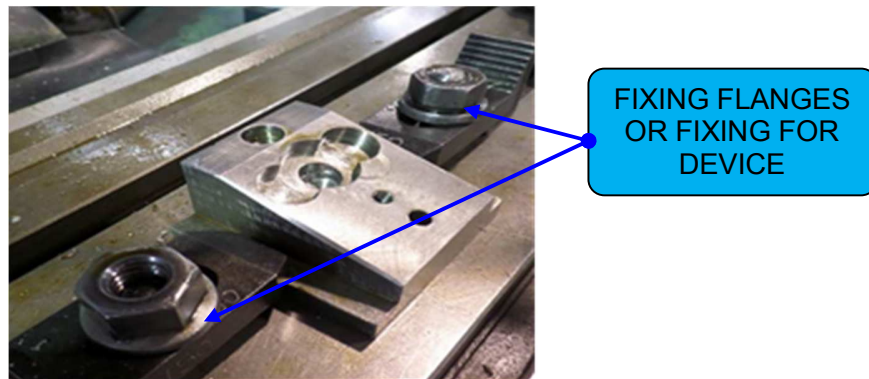
Sectional View

- Mounting device
The device is mounted on the mill table. (See Picture 5)



Picture 5

- Fixing of device
Fix device to the workbench. (See Picture 6)



Picture 6

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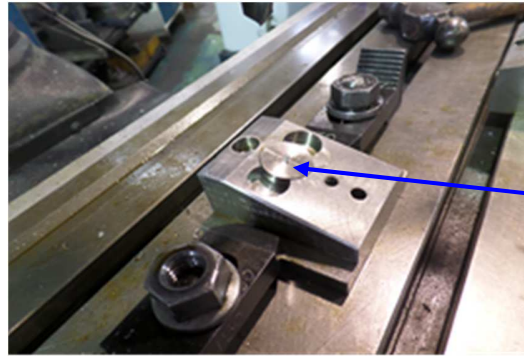
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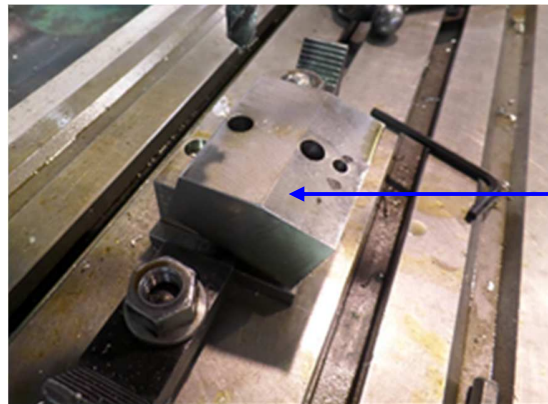
- Mounting of insert
Mounting the blank inserts on the device. (See Picture 7)



MOUNTING OF
BLANK INSERT

Picture 7

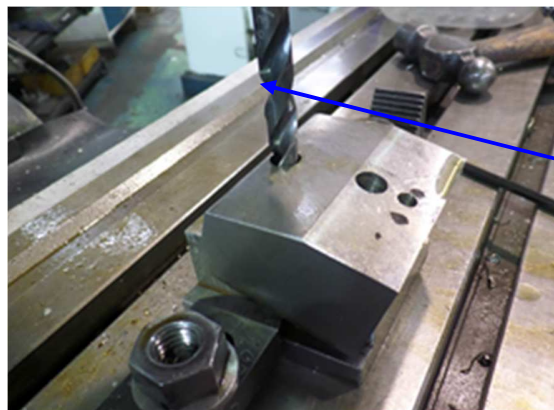
- Assembly
Assembly or mounting of centering device. (See Picture 8)



COVER AND
CENTERING
FOR DRILL

Picture 8

- Drilling
Perform drilling of the inserts. (See Picture 9)



ROCK DRILL
INSERT

Picture 9

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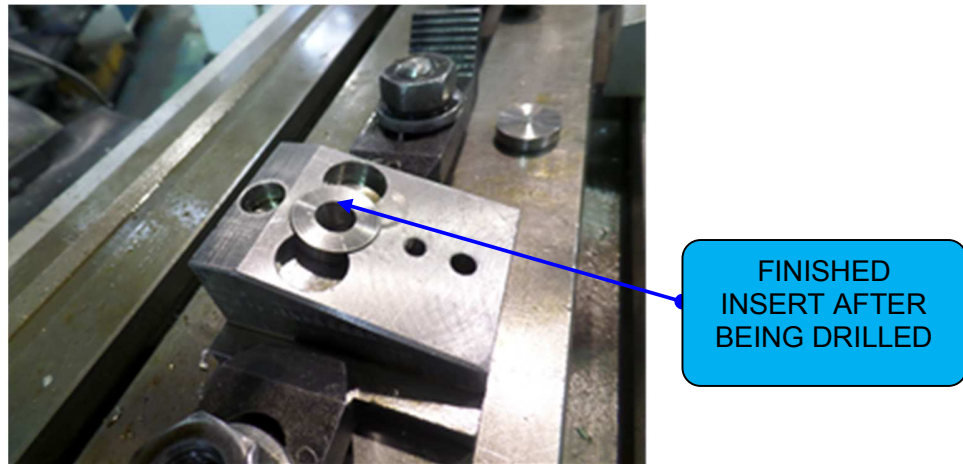
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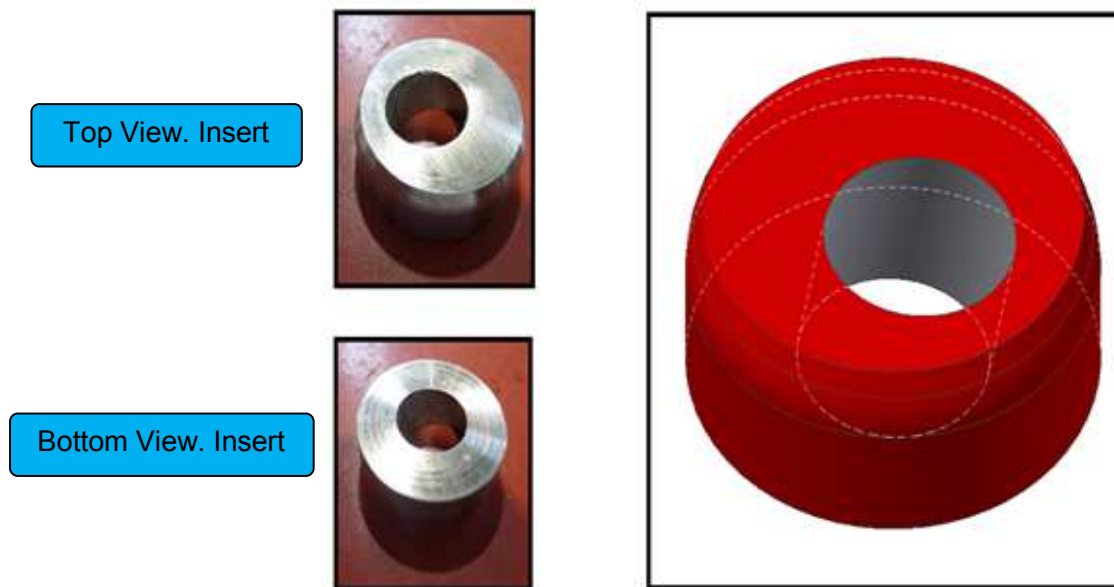
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- Disassembly
Removing cap. (See Picture 10)



Picture 10

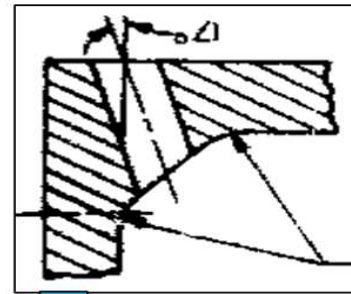
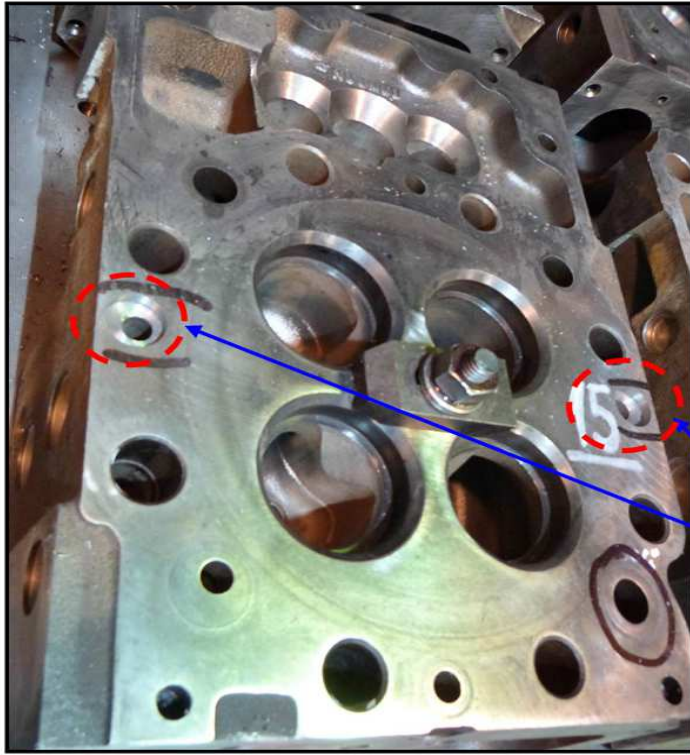
- Product
Insert finished and ready to be installed is obtained, which are sent to the store to have them as stock.



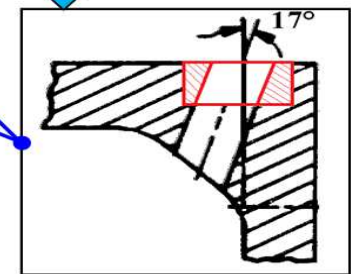
Picture 11

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- Installation of insert in Head 3500

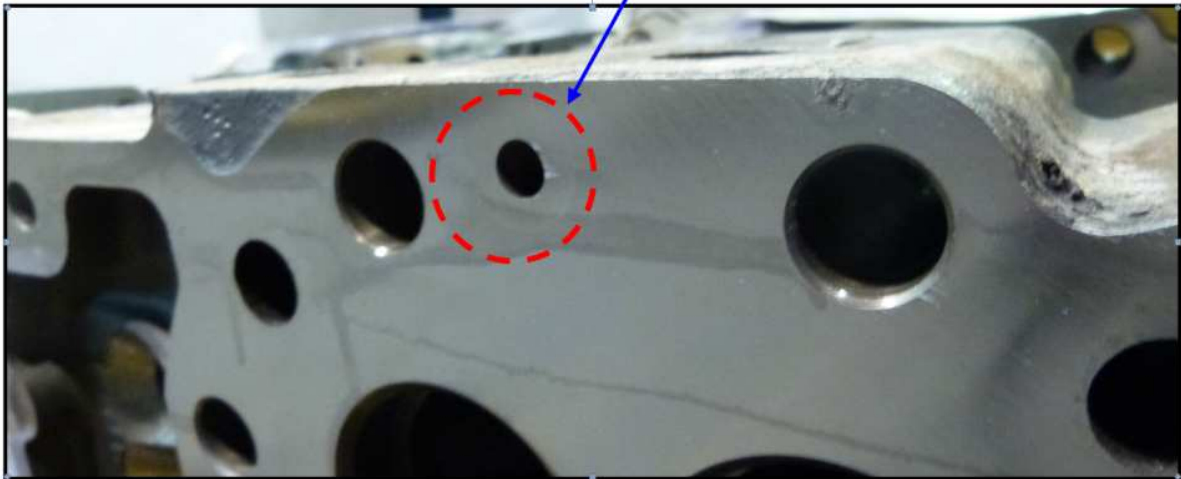


PUT IN INSERT IN
THE INCLINED
HOLE



Picture 12

INSERT, INSTALLED
AND RECTIFIED



Picture 13

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4.0 Benefits

- The drilling of the cylinder head was removed on the drill, transport, assembly and disassembly. Total time reduction is 39.29%

Old Process	Time	New Precess	Time
Conditioned accommodation.	0.2	Conditioned accommodation.	1.483
Metal Spray	1.543	Installing hairpiece.	0.05
Surface machining.	0.856	Grinding surface.	0.93
Surface cleaning.	0.527		
Grinding surface.	0.931		
	4.057		2.463
Time reduction.		39.29%	

- Standardization of inserts inventory to be applicable to heads 3500. (See picture 3)

5.0 Resources Required

Machine for head's inserts

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

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 Rafael Volta / Machine Chief
 Romulo Olarte / Machine Supervisor

Sponsor: Luis Aguilar / Manager Machine Shop

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